REPORT

Here is a brief description of each video in the series:

1) Introduction to Explainable AI:

This video provides an overview of the concept of explainable AI, which refers to the ability of machine learning models to provide understandable and transparent explanations for their decisions. The video discusses the importance of explainability in ensuring the reliability and trustworthiness of AI systems, as well as the challenges involved in achieving explainability.

2) Interpreting Machine Learning Models:

This video explores techniques for interpreting machine learning models, including feature importance analysis and partial dependence plots. It discusses how these techniques can be used to understand how models make decisions and identify potential biases.

3) Local Explanations with LIME:

This video introduces the Local Interpretable Model-Agnostic Explanations (LIME) technique, which is used to provide explanations for individual predictions made by machine learning models. The video explains how LIME works and demonstrates how it can be used to provide insight into the decision-making process of a model.

4) Global Explanations with SHAP:

This video discusses the Shapley Additive Explanations (SHAP) technique, which is used to provide global explanations for machine learning models. The video explains how SHAP works and how it can be used to identify the most important features in a model and understand its overall decision-making process.

5) Counterfactual Explanations and Adversarial Attacks:

This video explores the concept of counterfactual explanations, which are alternative scenarios that could have resulted in a different outcome. The video discusses how counterfactual explanations can be used to detect and defend against adversarial attacks, which are malicious attempts to manipulate the input data in a way that causes the model to make incorrect decisions.

6) Layer wise Relevance Propagation with MRI data:

This video discusses the application of the Layer wise Relevance Propagation (LRP) technique in analyzing magnetic resonance imaging (MRI) data to understand brain activity. The video explains how LRP can be used to generate relevance maps that highlight the regions of the brain that are most relevant to specific tasks or stimuli. It also demonstrates the use of LRP in a real-world example involving the analysis of brain activity in response to visual stimuli. From the video, one can learn about the LRP technique and its application in the field of neuroscience to understand brain activity and the underlying mechanisms of cognitive processes.

7) How to explain Graph Neural Networks (with XAI):

This video discusses the challenges of explaining the decisions made by Graph Neural Networks (GNNs) and presents techniques for achieving explainability using explainable AI (XAI) methods. The video

explains how GNNs are used for tasks such as node classification and link prediction in graphs and the importance of understanding how they arrive at their decisions. It then introduces several XAI techniques that can be used to generate explanations for GNNs. From the video, one can learn about the challenges of explaining GNNs and the techniques used to achieve explainability using XAI methods.

8) Explaining Twitch Predictions with GNNExplainer:

GNNExplainer, a graph neural network (GNN) explainability tool, can be used to understand the decision-making process of Twitch's prediction model. The video starts by introducing Twitch's prediction model, which uses a GNN to predict the outcome of live streams. The video then presents GNNExplainer, a tool that can explain how GNNs arrive at their decisions. The video demonstrates how GNNExplainer can be used to generate explanations for Twitch's prediction model and highlights the importance of interpretability in building trustworthy machine learning models. From the video, one can learn about the importance of interpretability in machine learning models and how GNNExplainer can be used to explain GNNs' decision-making processes.

9) How to handle Uncertainty in Deep Learning #1.1:

This video discusses the importance of modeling uncertainty in deep learning and presents different methods for achieving uncertainty estimation. The video introduces the concept of uncertainty in deep learning, which arises due to several factors such as limited data, model complexity, and measurement errors. The video then delves into various methods for estimating uncertainty in deep learning, such as Monte Carlo Dropout, Bayesian neural networks, and Variational Autoencoders (VAEs). The video also presents a real-world example that demonstrates how uncertainty estimation can be used for anomaly detection in medical images.

10) How to handle Uncertainty in Deep Learning #1.2:

This video discusses the concept of Monte Carlo Dropout (MCD) and its use for estimating uncertainty in deep learning models. The video introduces MCD, which is a technique for performing approximate Bayesian inference in deep learning models. The video demonstrates how MCD can be used for uncertainty estimation in image classification tasks and provides a comparison with other methods. The video also discusses the limitations and challenges of using MCD, such as computational overhead and model architecture.

11) How to handle Uncertainty in Deep Learning #2.1:

This video discusses the concept of epistemic and aleatoric uncertainty in deep learning and presents different methods for estimating them. The video defines epistemic and aleatoric uncertainty, which are two types of uncertainty in deep learning models. The video then delves into several methods for estimating epistemic and aleatoric uncertainty, such as Ensemble methods, Monte Carlo Dropout, and Maximum Likelihood Estimation (MLE). The video also provides a real-world example that showcases the application of these methods for estimating epistemic and aleatoric uncertainty in a medical imaging dataset.

12) How to handle Uncertainty in Deep Learning #2.2:

This video discusses the concept of deep ensembles and their use for uncertainty estimation in deep learning models. The video introduces the concept of deep ensembles, which involves training multiple deep neural networks on the same task and then combining their predictions to obtain an estimate of uncertainty. The video demonstrates how deep ensembles can be used for uncertainty estimation in image classification tasks and provides a comparison with other methods. The video also discusses the limitations and challenges of using deep ensembles, such as computational overhead and training time.

From these videos, one can learn about the importance of modeling uncertainty in deep learning, the different methods used for estimating uncertainty, including their strengths and limitations, and their applications in real-world scenarios such as medical imaging.

13) Causality and (Graph) Neural Networks:

This video discusses the concept of causality and its importance in machine learning models. The speaker explains how understanding causal relationships can help improve the interpretability and robustness of machine learning models, particularly in the context of graph neural networks. The video also discusses various methods and frameworks for inferring causality in machine learning models, such as GNN and SCM. From the video, viewers can learn about the significance of causal inference in machine learning and how it can be applied to graph neural networks.